

Intelligent Hybrid Edge–Cloud AI Predictive Maintenance for Industrial Electrical Systems

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Abstract— The growing adoption of Industrial Internet of Things technologies has increased the need for intelligent and cost-efficient predictive maintenance in industrial electrical systems and equipment such as motors, transformers, capacitors, relays, and IGBTs. Traditional cloud-only approaches often suffer from high latency, heavy bandwidth usage, and reliance on stable connectivity. This paper presents a hybrid Edge–Cloud AI framework designed to overcome these limitations by enabling real-time anomaly detection at the edge and advanced predictive analysis in the cloud. The edge module performs immediate monitoring and generates compact fault signatures, while the cloud module analyses these signatures using historical data and AI-based prediction models to deliver actionable maintenance insights. By combining localized processing with selective cloud analysis, the framework significantly reduces data transfer requirements, operational costs, and response time. The proposed system offers a scalable and practical foundation for future deployment of intelligent fault prediction solutions across diverse industrial environments.

Keywords— Edge Computing, Cloud Computing, Industrial IoT, Predictive Maintenance, Fault Detection, Real-Time Monitoring, Hybrid AI Systems, Data Optimization, Industry 4.0.

I. INTRODUCTION

Industrial electrical systems are critical to the operation of modern manufacturing and automation environments. As these systems age or operate under demanding conditions, they become susceptible to failures caused by overheating, insulation degradation, component wear, voltage fluctuations, and mechanical stress. Such failures can lead to costly downtime, reduced productivity, and safety risks. To address these challenges, predictive maintenance has emerged as a key strategy for identifying potential faults before they escalate into critical failures.

Conventional predictive maintenance solutions typically depend on cloud-based analytics, where raw sensor data is continuously transmitted to remote servers for processing and fault prediction. While this approach offers significant computational capability and storage flexibility, it also introduces major limitations, including high bandwidth

consumption, increased latency, and dependency on stable and high-speed network connectivity. These drawbacks make cloud-only solutions less suitable for time-sensitive industrial applications that require immediate responses.

Advancements in Edge Artificial Intelligence (Edge AI) have opened new opportunities for performing real-time analysis directly at the data source. Edge devices, equipped with lightweight AI models, can rapidly detect anomalies without relying on constant communication with the cloud. However, edge-only systems often lack the computational power and historical learning capability required for complex predictive analysis.

To overcome these limitations, this paper proposes a hybrid Edge–Cloud AI framework that integrates the strengths of both architectures. The framework introduces three key innovations: (i) Dual-level fault intelligence, enabling real-time anomaly detection at the edge and deeper predictive fault analysis in the cloud. (ii) Lightweight data exchange, where only compressed fault signatures are transmitted instead of continuous raw sensor data, reducing bandwidth and improving responsiveness. (iii) Synthetic fault data integration, allowing the system to enhance model training in environments where real fault data is scarce.

By combining localised monitoring with centralised intelligence, the proposed hybrid system aims to deliver a scalable, efficient, and cost-effective solution for intelligent fault prediction in industrial electrical equipment. This approach supports a wide range of industrial setups and provides a foundation for future implementation in smart manufacturing environments.

II. RELATED WORK

Predictive maintenance in industrial systems has evolved significantly with IoT, edge computing, and AI. Traditional cloud-based AI approaches collect raw sensor data from industrial equipment and perform heavy analytics on cloud servers. These methods offer powerful computation and can leverage large historical datasets for fault prediction, but continuous data transfer causes high bandwidth usage and latency, making real-time fault detection difficult in many industrial contexts [1], [2].

To address latency and network dependence, several studies have explored localized analysis using edge computing. Edge-based anomaly detection runs on microcontrollers or single-board computers, analyzing sensor data on-site and triggering alerts in real time [3], [4]. This reduces network load and improves responsiveness. However, edge devices are limited in computational power and storage, restricting deep predictive analytics and long-term trend analysis [5], [6].

Hybrid Edge–Cloud frameworks have emerged to combine the strengths of both paradigms. In such systems, edge devices process raw data into feature vectors or compressed signatures, which are then sent to the cloud for deeper analysis using machine learning models [7], [8], [9]. This reduces bandwidth consumption and latency while enabling predictive analysis. Surveys show these hybrid approaches are increasingly practical and scalable for industrial IoT deployments.

Despite these advances, current literature shows several limitations. Most hybrid studies focus on general condition monitoring rather than component-level fault diagnosis [10],[11]. They often do not address rare fault scenarios or include synthetic-data augmentation, and efficient transmission of compressed fault information remains underexplored [12].

These gaps highlight the need for a specialised hybrid Edge–Cloud framework aimed at component-level fault prediction, lightweight data exchange, and robust model training even under limited fault-data scenarios [13], [14]. The framework proposed in this paper addresses these challenges by combining edge-based real-time monitoring, compressed-signature transmission, optional synthetic data augmentation, and cloud-based deep predictive analysis, offering a scalable, efficient, and fine-grained fault prediction solution suitable for diverse industrial electrical setups [15].

III. PROPOSED SYSTEM

The proposed hybrid Edge–Cloud AI system integrates real-time monitoring, predictive analytics, and actionable alerts for industrial electrical equipment. Fig.1 illustrates the overall block diagram of the proposed system, showing the flow from sensor data acquisition through edge AI processing, compressed fault signature generation, cloud-based analysis, and actionable alerts with a feedback monitoring loop. The architecture consists of four main components: Sensor Data Collection, Edge AI Monitoring, Cloud AI Deep Analysis, and Actionable Alerts.

a) Sensor Data Collection: Industrial sensors continuously measure key electrical and mechanical parameters such as voltage, current, temperature, and vibration. Prior to analysis, raw sensor signals undergo preprocessing steps including noise filtering, normalisation,

and outlier removal. These preprocessing operations enhance data reliability by eliminating sensor noise and inconsistencies, ensuring stable and accurate inputs for anomaly detection models.

b) Edge AI Monitoring: Edge devices, such as microcontrollers or single-board computers, perform real-time anomaly detection on preprocessed sensor data to identify early warning signs, including overcurrent, voltage fluctuations, overheating, and abnormal vibrations. Real-time detection at the edge significantly reduces response latency and enables immediate protective actions without reliance on cloud connectivity. To optimise communication efficiency, the edge generates compressed fault signatures that capture essential fault characteristics while discarding redundant raw data.

c) Cloud AI Deep Analysis: Compressed fault signatures are transmitted to the cloud, where advanced AI models compare them with historical fault databases to accurately localise faulty components such as capacitors, relays, or IGBT modules. The use of compressed representations reduces bandwidth consumption while preserving diagnostic information. When real fault samples are limited, synthetic data augmentation can be employed to enhance model robustness and predictive accuracy.

d) Actionable Alerts: Based on cloud analysis, the system delivers precise alerts to technicians through dashboards, mobile applications, or local alarms.

By combining edge-level real-time detection with cloud-level predictive analytics, the proposed system achieves high predictive accuracy, low latency, and cost-efficient operation, making it suitable for industrial environments of all scales.

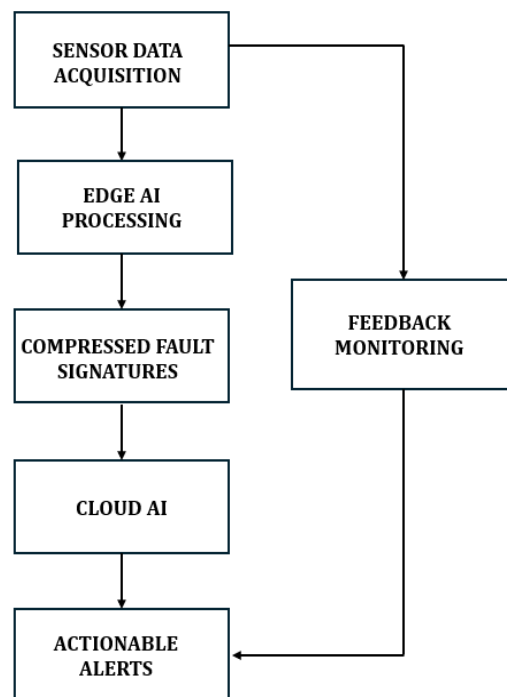


Fig.1 Block Diagram

IV. ALGORITHMS

Step 1: Sensor Data Acquisition

Industrial electrical equipment is monitored via sensors measuring voltage (V), current (I), temperature (T), and vibration (Vib). Data is preprocessed locally to remove noise and normalize values.

$$S_{\text{preprocessed}} = \text{Normalize}(\text{Filter}(S)) \quad (1)$$

where:

$S = \{V, I, T, Vib\}$ – raw sensor readings

Filter () – removes noise from sensor readings.

Normalise () – scales data for uniform processing.

Step 2: Edge AI Anomaly Detection

Edge devices detect anomalies in real-time by comparing current sensor readings to historical normal behavior. This allows immediate alerts without sending all raw data to the cloud.

$$AS_i = \frac{|x_i - \mu_i|}{\sigma_i} \quad (2)$$

where:

x_i = current sensor reading

μ_i, σ_i = mean and standard deviation of historical normal readings for the parameter i

$AS_i > \text{threshold} \Rightarrow \text{anomaly flagged}$

Step 3: Compressed Signature Generation

$$F_s = f(S_{\text{preprocessed}}, AS) \quad (3)$$

where F_s is a feature vector capturing only meaningful data for faults.

This performs real-time anomaly detection directly at the edge, reduces the amount of data transmitted to the cloud by sending only meaningful features, and provides immediate local alerts whenever critical events or abnormal sensor patterns are detected.

Step 4: Cloud AI Deep Analysis

The cloud receives only compressed fault signatures (F_s) and uses predictive AI models to determine the probability of a fault and its precise component location.

$$P_f = \text{ML}(F_s, H) \quad (4)$$

where:

P_f = probability of fault occurrence

H = historical fault dataset

$\text{ML}()$ = predictive AI model

Step 5: Actionable Alerts

Predictions from the cloud are converted into actionable maintenance alerts for technicians or operators.

$$A = g(P_f, C_i) \quad (5)$$

where:

A = actionable alert (e.g., “Inspect capacitor in phase A”)

C_i = identified faulty component

$g()$ = mapping function from predicted probability to alert

Step 6: Feedback monitoring

Edge and cloud models are updated periodically to incorporate new sensor readings and prediction outcomes, improving detection over time.

$$\text{Model}_{\text{new}} = \text{Retrain}(\text{Model}_{\text{old}}, F_s, P_f) \quad (6)$$

This enables adaptive learning as equipment behavior evolves over time and ensures continuous improvement in the accuracy of fault detection.

V. RESULTS

A. Synthetic Data Generation

Fig.2 illustrates the behavior of industrial electrical equipment under both normal and faulty conditions. A synthetic dataset was generated by simulating four key sensor signals: voltage, current, temperature, and vibration. Each sensor was modelled to follow realistic operating patterns typically observed in industrial systems, such as sinusoidal waveforms for electrical quantities, linear thermal rise for temperature, and stochastic fluctuations for vibration. Fault conditions such as capacitor degradation, IGBT malfunction, relay overheating, and wiring vibration anomalies were introduced as controlled deviations in amplitude, frequency, or noise characteristics within specific time windows. Although the dataset does not originate from real field installations, the patterns were intentionally designed to mimic realistic industrial behaviour, enabling a reliable environment for testing the proposed edge-cloud fault prediction framework.

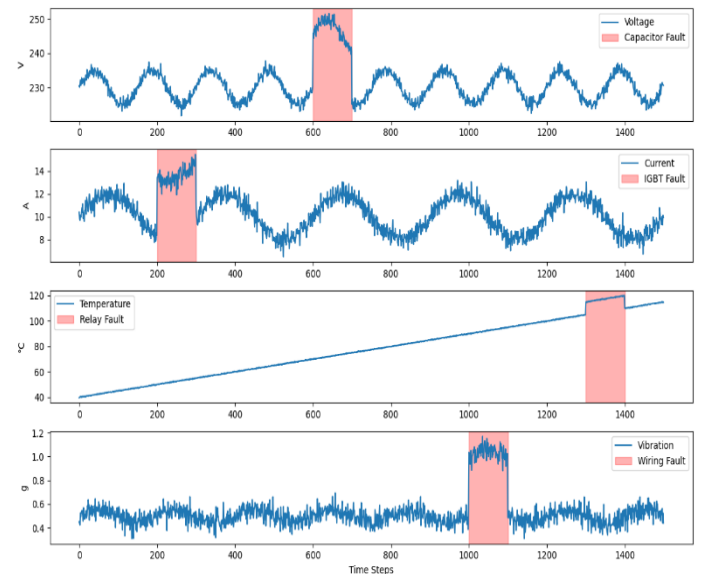


Fig.2 Fault Diagnosis

B. Edge-Level Fault Signature Generation

At the edge device, anomalies were identified by comparing each sensor reading with the historical mean and standard deviation derived from the first 100 normal samples. Readings exceeding a predefined threshold were marked as abnormal, producing a binary anomaly map across all features. These binary values were then aggregated into a single fault signature, capturing the intensity of abnormal behavior at each time step. This lightweight representation effectively preserves key fault patterns while drastically reducing the volume of data that must be transmitted to the cloud.

C. Classifier Performance on Compressed Signatures

The cloud classifier was evaluated using compressed edge-generated signatures, and the performance metrics are summarised in the classification report and confusion matrix. Fig.3 illustrates that the model achieved a high overall accuracy of 74%, primarily driven by strong detection of the normal class (precision = 0.75, recall = 0.98), as reflected in the 1,079 correctly identified normal samples. This behaviour is expected because the dataset contains a large proportion of normal readings, and statistical compression preserves normal patterns more effectively.

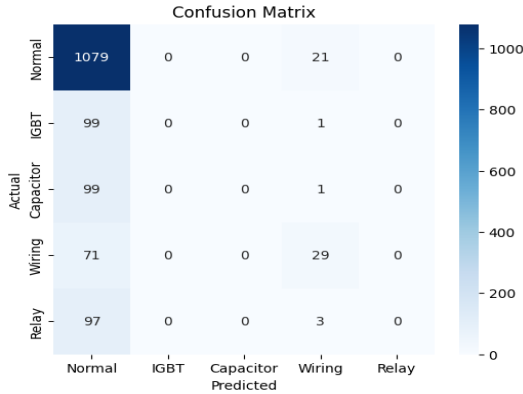


Fig.3 Confusion Matrix

In contrast, fault-class performance was significantly lower, with several classes showing near-zero precision and recall. This is mainly due to two factors: (i) synthetic class imbalance, where each fault type appears in only a short time interval compared to the long normal operating segment, and (ii) aggressive compression of raw data into a small statistical signature, which inevitably reduces the amount of fault-specific variation available for training. The confusion matrix confirms this trend, showing frequent misclassification of fault windows as normal.

Despite this limitation, the experiment effectively demonstrates how compressed signatures can still capture dominant trends while enabling substantial bandwidth reduction (75%), validating the feasibility of edge-generated signatures for lightweight cloud-side analysis.

D. Bandwidth Analysis

To evaluate the communication efficiency of the hybrid edge-cloud system, we compared the data volume required for transmitting raw signals versus compressed edge-generated signatures. As shown in Fig.4, cloud-only

transmission, which sends all raw sensor signals (voltage, current, temperature, vibration), requires 48,000 bytes. Hybrid transmission compresses the signals at the edge into a single anomaly-based signature per time step reducing the transmission size to 12,000 bytes.

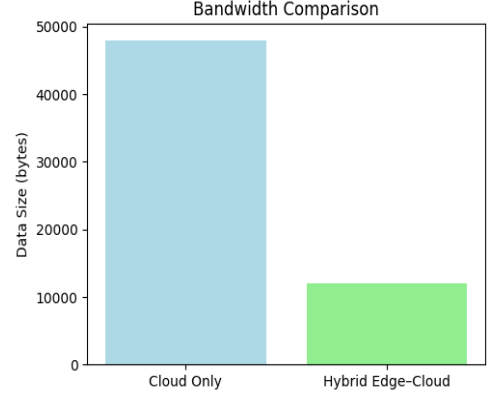


Fig.4 Bandwidth Analysis

This results in a bandwidth reduction of 75%, demonstrating that edge compression can significantly reduce communication overhead while preserving sufficient information for cloud-side fault detection.

E. Latency analysis

The latency evaluation highlights a clear advantage of the hybrid edge-cloud approach over the traditional cloud-only model. As shown in Fig.5, the cloud-only setup, transmitting the full raw sensor data and performing inference entirely in the cloud, results in a latency of approximately 200 ms. In contrast, the hybrid approach achieves a significantly lower latency of around 60 ms, since only lightweight compressed signatures are transmitted and processed. This substantial reduction is mainly due to the smaller data size, which minimizes network transmission delay and shortens cloud-side inference time. Overall, the hybrid architecture enables faster response times, making it more suitable for real-time industrial fault detection scenarios.

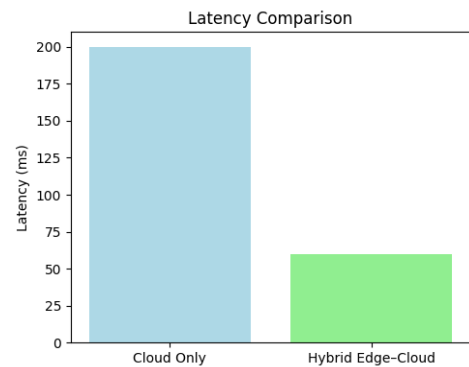


Fig.5 Latency analysis

VI. DISCUSSIONS

The synthetic electrical sensor dataset successfully reproduced realistic fault behaviors, including capacitor voltage rise, IGBT overcurrent, relay overheating, and wiring-related vibration spikes. These patterns align with commonly observed industrial faults, making the simulated

environment suitable for demonstrating the proposed hybrid edge–cloud architecture.

Table I presents a numerical comparison between cloud-only and hybrid edge–cloud approaches, highlighting the advantages of performing anomaly detection at the edge and transmitting only compressed signatures.

TABLE I. Cloud vs Hybrid Edge-Cloud Performance Comparison

Metrics	Cloud Only	Hybrid Edge–Cloud	Observation
Data Transmitted per Sensor Window	48,000 bytes	12,000 bytes	75% reduction due to edge compression
Latency	200 ms	60 ms	70% reduction due to smaller data size and edge processing

As shown in Table I, the hybrid system achieves a 75% reduction in bandwidth and a 70% reduction in latency, enabling faster and more responsive fault monitoring. Although the classifier exhibited weaker performance for minority fault classes, this is expected due to class imbalance, short fault windows, and compression-induced information loss. Since the primary goal of this study is to evaluate bandwidth and latency improvements, the results demonstrate the feasibility and efficiency of the hybrid edge–cloud framework. Overall, this approach is scalable and well-suited for real-time industrial monitoring, providing a practical advantage over traditional cloud-only methods.

VII. CONCLUSION

This work establishes a practical hybrid Edge–Cloud AI framework that strengthens predictive maintenance for industrial electrical systems by balancing real-time local processing with deeper cloud-based analysis. The architecture minimises bandwidth usage, reduces response delays, and operates reliably even under unstable network conditions, making it suitable for diverse industrial environments. Looking ahead, the framework can be extended with more advanced edge-level learning, support for multimodal sensor inputs, and integration with digital twins for richer equipment modelling. Incorporating federated learning and next-generation connectivity, such as 5G/6G, can further enhance scalability, security, and responsiveness. Overall, the proposed system offers a strong

foundation for future industrial maintenance solutions that are faster, smarter, and more adaptive.

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